



Sanat Agrawal
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Communication & Signal Processing

22B3919
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 18/05/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2027	9.54 Core: 204 Total: 240
Intermediate	C.B.S.E	Arihant Public School	2022	96.20%
Matriculation	I.C.S.E	Little Flower School	2020	97.83%

Pursuing a **Minor Degree** offered by **Centre for Machine Intelligence and Data Science (C-MInDS)**

SCHOLASTIC ACHIEVEMENTS

- Co-first authored **2 research papers** submitted to **ACL 2026** and **IEEE SLT 2026** as an undergraduate ('26)
- Scored a perfect **10/10 Semester Performance Index (SPI)** in the Spring 2025 semester at IIT Bombay ('25)
- Awarded **Heyning-Roelli Scholarship** to pursue an academic exchange semester at **EPFL, Switzerland** ('25)
- Selected globally for fully-funded **Summer Research Initiative (SURI)** at **Arizona State University** ('24)

TECHNICAL EXPERIENCE

Undergrad Researcher

(Mar '26 - May '26)

Digital Audio Processing Lab, IIT Bombay | Guide: Prof. Srikanth Raj Chetupalli

- Built **Indic-CLAP**, extending **CLAP** text-audio retrieval to **9 Indic languages** via cross-lingual distillation
- Trained an **XLM-RoBERTa** on **GPT-4o-mini** translated **AudioCaps** captions, tuning only **394K** parameters
- Proposed a **hybrid distillation** objective, surpassing the English **CLAP** teacher by **+14.3 R@1** in retrieval tasks
- Traced gains to a reduced **modality gap** via centroid analysis and **co-authored** paper submitted to **ACL 2026**

Visiting Student Researcher

(Oct '25 - Feb '26)

Idiap Research Institute, Switzerland | Guide: Dr. Mathew Magimai-Doss

- Proposed **Concatenative Neural Speech Synthesis**, fusing **Festival** unit selection with **WavLM** SSL features
- Built a **direct feature-concatenation** pipeline with **kNN-VC** and **HiFi-GAN**, skipping waveform re-encoding
- Delivered **zero-shot synthesis** method to **40 LibriSpeech** target speakers using only **0.9 hrs** of source speech
- Gained **+0.81 UTMOS** over Festival baseline across **7 voices**; **co-authored** paper submitted to **IEEE SLT 2026**

Research Intern

(May '24 - July '24)

Arizona State University, USA | Guide: Prof. Ayan Mallik

- Built an **all-loss-inclusive** mathematical model of DAB converter in **MATLAB** that emulates the **real converter**
- Implemented a Physics Informed **Neural Network** to estimate **parametric uncertainties** upto a **92% accuracy**
- Utilized **adaptive loss-optimal control** in converter, reducing **power losses by 67%** under varying conditions
- Collected **Data points** from the actual **hardware setup** of converter for validation of machine learning model

WORK EXPERIENCE

Founding Engineer

(Aug '25 - Present)

Novare Talent | SINE, IIT Bombay, Mumbai | Backed by NVIDIA Inception

- Built **3 production products** on **Next.js/FastAPI/Supabase** serving **2.3K+ profiles** across **14 IIT** campuses
- Engineered **Arena X**, a real-time **1v1** coding arena with **ELO** tier matchmaking over **sandboxed** test-case judging
- Built **vector** resume-JD **semantic search** engine with **LLM** reranking, cutting recruiter screening time to **hours**
- Scaled to an official **IIT Bombay Alumni Association** partnership, placing hires at **series A** startups and firms

AI & Systems Intern

(June '25 - July '25)

SalesAgent.ai | Mumbai

- Integrated **OpenAI Whisper** for **real-time** streaming and **multilingual speech-to-text (STT)** support
- Enhanced transcription latency and accuracy by incorporating **Sarvam Ai STT** into the hybrid STT architecture
- Worked on optimizing the end-to-end voice agent flow, from live audio input to voice output, ensuring scalability

KEY PROJECTS

Autonomous Racing Agent via Reinforcement Learning

(April '25 - May '25)

Course Project | Foundations of Intelligent and Learning Agents | Instructor: Prof. Shivaram Kalyanakrishnan

- Developed an **autonomous agent** to navigate dynamic racing tracks using **13x13 grid** via **CMA-ES** over **100 generations** with a **30-agent population**, engineering **3 core features** that cut input dimensionality by **>95%**
- Designed a **hybrid policy** achieving **3.2x** higher distance and **>80%** lower crash rate over baseline strategies

LLM Decoding and Inference Acceleration with Medusa

(April '25 - May '25)

Course Project | Advanced Machine Learning | Instructor: Prof. Sunita Sarawagi

- Implemented 4 **LLM decoding** strategies (Greedy, Top-k, Nucleus, Temperature) for **Hindi-to-English** translation using **Llama-2** on the **IN22-Gen dataset**, evaluated using BLEU and ROUGE scores for benchmarking
- Developed **trie-based** word-constrained and **Medusa-style** speculative decoding, achieving **3×** faster inference

Cough-Based Disease Detection using ML

(Jan '25 - May '25)

RnD Project | DH-307 | Guide: Prof. Nirmal Punjabi

- Engineered a dual-input neural architecture (**FFM**) combining **BiLSTM** and **CNN** models, achieving **96.33%** accuracy in **Tuberculosis** detection and **94%** accuracy in **COVID-19** classification from cough audio signals
- Improved F1-score by **14%** on imbalanced datasets (**Coughvid**, **Coswara**) using **SMOTE** and **RUS** techniques
- Identified key features via **MRMR** and **Kolmogorov–Smirnov** tests, reducing the feature set by **89%** (193→17)

AlphaSense AI: Live Equity Trading System

(April '26 - Aug '26)

Self project | Selected for **IITB-Groww INV.ENT** Program | DSSE, IIT Bombay

- Built an event-driven signal engine fusing **FinBERT/LLM** news sentiment, **XBRL** fundamentals and Z-scores
- Deployed unattended on **AWS EC2** with **cron**, **systemd** and **Streamlit**, executing live orders via broker API
- Instrumented a nightly feedback loop tracking rolling Sharpe and win-rate drift to auto-flag parameter decay

Numerical Methods for Visual Computing & ML

(Sept '25 - Dec '25)

Course Project | CS-328 | EPFL, Switzerland

- Built a **reverse-mode Auto Differentiation** engine to train neural networks without external ML frameworks
- Implemented **SVD & PCA** for facial recognition, achieving dimensionality reduction on high-res image datasets
- Developed Least Squares solvers using **QR factorization** to perform image deblurring and mesh deformations

SEMESTER EXCHANGE

École Polytechnique Fédérale de Lausanne (EPFL) | Grade - 5.61/6

(Sept '25 - Jan '26)

- Completed a semester exchange at **EPFL**, ranked **22nd globally** (QS 2026), on a **fully funded Scholarship**
- Studied Quantum Information Processing, Numerical Methods for ML and Nanosystem Modelling in **COMSOL**
- Collaborated with peers from **40+ nationalities**, adapting to a French-speaking academic and cultural setting

POSITIONS OF RESPONSIBILITY

Web Coordinator | Student Alumni Relations Cell, IIT Bombay

(Jun '23 - April '24)

- Working with a team of **80+ members** organizing **40+ events** to foster relations among **60k+ alumni** and students
- Developed a **registration portal** for Mentors & Mentees of ASMP using HTML, CSS, JS, **Bootstraps & Django**

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL, VHDL, Bash
ML & Data	PyTorch, HuggingFace, Scikit-learn, TensorFlow, Keras, NumPy, Pandas, OpenCV
Development	FastAPI, Next.js, React, Supabase, PostgreSQL, pgvector, Node.js, HTML/CSS
Systems & Tools	AWS EC2, Docker, Git, Linux, CUDA, Slurm, systemd, nginx, Streamlit, Jupyter
Engineering	MATLAB, Simulink, COMSOL, Quartus Prime, LTspice, KiCad, Fusion 360

KEY COURSES UNDERTAKEN

ML & Data Science	Advanced Machine Learning, Foundations of Intelligent and Learning Agents, Automatic Speech Processing, Advanced Image Processing, Data Science & ML
Mathematics	Linear Algebra, Calculus, Differential Equations, Probability & Random Processes, Markov Chains & Queuing Systems, Online Learning & Optimisation
Systems & Signals	Microprocessors, Digital Systems, Signal Processing, Digital Signal Processing, Error Correcting Codes, Communication Networks, Advanced Data Networks
Others	Fintech Entrepreneurship, Business Fundamentals for Technopreneurs, Economics

EXTRACURRICULAR ACTIVITIES

- Conducted an **8-session** AI literacy workshop, teaching **15+ school students** basic AI concepts and tools (2026)
- Backpacked across **15+ European countries** during the semester exchange, planning travel independently (2025)
- Completed a yearlong intensive training programme in **chess** under **NSO**, IIT Bombay (2023)
- Conducted **doubt solving** sessions for underprivileged students in association with **IIT Kanpur** (2023)